

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think>

- The patient is diagnosed with community-acquired pneumonia and treated with cephalexin, which is an antibiotic. This suggests a bacterial infection.
- Cephalexin is effective against Streptococcus pneumoniae, which is a common cause of community-acquired pneumonia.
- Streptococcus pneumoniae is a gram-positive bacterium.
- Gram-positive bacteria have peptidoglycan as a major component of their cell wall.
- The immune response to gram-positive bacteria typically involves the activation of TLR2 and TLR5 on the surface of macrophages, not TLR1.
- The Th1 response is more associated with intracellular pathogens, not the common community-acquired pneumonia.
- Encapsulation is a mechanism used by some pathogens to evade the immune response, but it is not a primary response for Streptococcus pneumoniae.

Based on this reasoning, the best description of the immune response elicited by the pathogen causing this patient's condition is that peptidoglycan is its major antigen that induces an immune response.

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<answer>C. Peptidoglycan is its major antigen that induces an immune response.</answer>